HYBRID DEEP LEARNING FRAMEWORK FOR ADAPTIVE LINK DETECTION AND WEB THREAT APPLICATION

B.Tech Major Project Synopsis — 2026–2027

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Abstract

The rapid growth of internet usage has increased cyber threats such as phishing, malware, and fraudulent websites. Traditional security systems mainly rely on blacklist and signature-based detection, which are ineffective against newly created malicious URLs. This project proposes a **Hybrid Deep Learning Framework for Adaptive Malicious Link Detection and Web Threat Intelligence** to identify malicious URLs before users access them. The system analyzes URL features using deep learning techniques to classify links as safe or malicious. It provides real-time threat detection, warning alerts, and improves user cybersecurity awareness by helping users browse the web safely.

Problem Statement

Existing malicious URL detection systems mainly rely on blacklist databases and signature-based techniques, which are unable to detect newly created or zero-day malicious websites. As cyber threats continue to evolve, these traditional methods become less effective in preventing phishing, malware, and fraudulent attacks. Therefore, there is a need for an intelligent Hybrid Deep Learning-based system that can analyze URL features, accurately identify malicious links in real time, and enhance user protection against online threats.

Objectives

- Develop a Hybrid Deep Learning model to detect malicious URLs.
- Identify phishing, malware, and fraudulent websites with high accuracy.
- Extract and analyze URL features for effective threat detection.
- Provide real-time prediction and warning alerts for malicious links.
- Improve cybersecurity awareness and promote safe web browsing.

System Architecture

URL Input → Feature Extraction → Data Preprocessing → Machine Learning Training → Threat Prediction Engine → Risk Classification → Alert & Reporting System

Module 1: Dataset Collection Module

- Collects legitimate and phishing URLs from trusted datasets.
- Ensures balanced and high-quality training data.
- Stores labeled URLs for model development**.**

Module 2 : Feature Extraction Module

- Extracts important URL characteristics.
- Identifies features like URL length, HTTPS, IP address, and special characters.
- Converts URLs into numerical features for machine learning.

Module 3: Data Preprocessing Module

- Removes missing or duplicate data.
- Cleans and normalizes the dataset.
- Splits data into training and testing sets.

Module 4: Model Training Module

- Trains multiple machine learning algorithms.
- Compares model performance using evaluation metrics.
- Selects the best-performing model for prediction.

Module 5: Threat Prediction Module

- Classifies URLs as safe or malicious.
- Uses the trained model for real-time prediction.
- Provides fast and accurate detection results.

Module 6: Alert & Reporting Module

- Displays warning messages for malicious URLs.
- Generates prediction results and security reports.
- Helps users make safe browsing decisions.

Technology Stack

Component	Technology
Programming Language	Python
Frontend	HTML, CSS, JavaScript
Backend	Flask
Database	MySQL
Libraries	Pandas, NumPy, Scikit-Learn, Matplotlib
Tools	VS Code, Jupyter Notebook

Expected Outcomes

- High accuracy URL classification.
- Real-time malicious link detection.
- Reduced phishing attacks.
- Improved cybersecurity awareness.
- Better protection against online threats.

Conclusion

Hybrid Deep Learning Framework For Adaptive Malicious Link Detection And Web Threat Intelligence offers a proactive cybersecurity solution capable of Detecting malicious URLs before users interact with them. By leveraging machine learning, the System enhances online safety and reduces the risk of phishing and malware attacks.

Technology Stack